

NLP Report

Multilingual Hate Speech Detection in Online Text

A Cross-Lingual Classification Study Using RoBERTa on English and Roman Urdu Corpora

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ABSTRACT

The proliferation of social media platforms has intensified the spread of abusive and hateful language, demanding robust automated detection systems. This paper presents a multilingual hate speech detection framework that jointly trains on English-language toxic comments and Roman Urdu hate speech. Two publicly available datasets are consolidated into a unified binary corpus of approximately 175,000 samples and used to fine-tune RoBERTa, a robustly optimised version of BERT. The combined dataset is partitioned using a 70/10/20 train-validation-test split, with class imbalance addressed through weighted binary cross-entropy loss. Evaluation on the held-out test set demonstrates the effectiveness of the proposed approach, with performance reported in terms of accuracy, F1-score, precision, and recall. The trained model is released publicly on the Hugging Face Hub for community use. This study contributes a reproducible, cross-lingual pipeline for hate speech detection and highlights persistent challenges such as code-switching and label ambiguity in low-resource language settings.

Keywords: hate speech detection, RoBERTa, multilingual NLP, Roman Urdu, text classification, transformer fine-tuning, toxic language

1. INTRODUCTION

The rapid expansion of user-generated content on platforms such as Twitter, Facebook, Reddit, and YouTube has created an unprecedented challenge for online governance. Harmful speech — including threats, insults, ethnic slurs, and incitement to violence — not only harms individual users but also erodes social cohesion and public discourse. Manual moderation at scale is economically infeasible, which has made automated content moderation an active and consequential research area within natural language processing (NLP).

While significant progress has been made for high-resource languages, particularly English, the detection of hateful content in low-resource and code-switched languages remains severely understudied. Roman Urdu — a transliteration of the Urdu language using Latin script — is widely used across South Asian social media communities yet lacks the rich annotated corpora and pre-trained models available for English. Extending hate speech detection to such languages is both a technical challenge and a social necessity.

This paper addresses both concerns. We construct a unified multilingual dataset by merging the Jigsaw Toxic Comment Classification dataset, a large-scale English corpus of Wikipedia talk-page comments, with a Roman Urdu hate speech corpus collected from social media platforms. Both datasets are harmonised into a common

binary schema (hate speech / non-hate speech) and used to fine-tune RoBERTa (Liu et al., 2019), a transformer-based language model that has demonstrated state-of-the-art performance on a wide range of NLP benchmarks.

The primary contributions of this work are as follows:

- (i) Construction of a merged, multilingual hate speech corpus spanning English and Roman Urdu.
- (ii) Fine-tuning of RoBERTa with an imbalance-aware loss function, documented with complete hyperparameters.
- (iii) Public release of the trained model on the Hugging Face Hub (nalain7/NLP_Hate_Speech_v2.0).
- (iv) Discussion of cross-lingual generalisation challenges and directions for future work.

The remainder of this paper is organised as follows. Section 2 reviews related literature. Section 3 describes the datasets employed. Section 4 details the methodology, including preprocessing, model architecture, and training procedure. Section 5 presents experimental results. Section 6 discusses findings and limitations. Section 7 concludes the paper.

2. RELATED WORK

Research into automated hate speech and toxicity detection has followed the broader trajectory of NLP, progressing from lexicon-based heuristics and classical machine learning to deep neural architectures and, most recently, large pre-trained language models.

2.1 Classical Approaches

Early work on abusive language detection relied on manually curated lexicons combined with bag-of-words or TF-IDF feature representations fed into logistic regression or support vector machine (SVM) classifiers. Nobata et al. (2016) demonstrated that a combination of character n-grams, syntactic features, and word embeddings substantially outperformed a comment-based lexicon baseline on Yahoo! Finance and News comments. Although computationally efficient, these approaches suffer from limited contextual understanding and poor generalisation to unseen vocabulary, particularly in colloquial or multilingual text.

2.2 Deep Learning Approaches

The introduction of recurrent neural networks — particularly Long Short-Term Memory (LSTM) networks and their bidirectional variants — enabled models to capture sequential dependencies in text. Badjatiya et al. (2017) showed that LSTM networks trained on character-level embeddings outperformed classical models on Twitter hate speech data. Convolutional neural networks (CNNs) were similarly applied to capture local n-gram features. However, both architectures require task-specific training from scratch and struggle with long-range dependencies.

2.3 Transformer-Based Models

The introduction of BERT (Bidirectional Encoder Representations from Transformers; Devlin et al., 2019) marked a paradigm shift in NLP. By pre-training on large unlabelled corpora with masked language modelling and next-sentence prediction objectives, BERT acquires rich contextual representations that transfer effectively to downstream classification tasks through fine-tuning. Mozafari et al. (2020) demonstrated that BERT fine-tuned on the Jigsaw dataset achieved significant improvements over prior state-of-the-art results. RoBERTa (Liu et al., 2019) improved upon BERT by removing the next-sentence prediction objective, training with larger batches and more data, and using dynamic masking, leading to more robust representations.

2.4 Base Paper: Bilal et al. (2023)

Title: Roman Urdu Hate Speech Detection Using Transformer-Based Model for Cyber Security Applications

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This paper constitutes the primary base reference for the present study by reason of its direct alignment with both the language domain (Roman Urdu) and the methodological approach (transformer-based hate speech classification). Bilal et al. address hate speech detection on social media platforms — specifically Twitter and Facebook — using Roman Urdu text, a low-resource language that receives virtually no coverage in mainstream NLP literature. Their work is the most directly comparable prior study to the present research and its findings both motivate and contextualise the design decisions made in this project.

The central contribution of Bilal et al. is the development of BERT-RU, the first pre-trained BERT model built from scratch for Roman Urdu text. BERT-RU is trained on a custom-built corpus of 173,714 Roman Urdu text messages assembled from diverse social media sources. The authors explicitly note that the BERT Multilingual Cased model, despite covering 104 languages, does not include Roman Urdu — a gap that severely limits the applicability of off-the-shelf multilingual transformers to this language. By pre-training a dedicated model, they demonstrate that domain-specific pretraining yields substantial improvements in classification performance for low-resource languages where general multilingual models fall short.

In addition to BERT-RU, the study evaluates several traditional deep learning baselines: LSTM, BiLSTM, BiLSTM with an Attention Layer, and CNN. These models are trained on the same Roman Urdu corpus and serve as performance reference points. The systematic comparison across these architectures allows the authors to quantify the performance gain attributable to transformer-based contextual representations over sequential and convolutional feature extractors.

The key findings of Bilal et al. that directly inform and motivate the design choices in the present work are as follows:

- (i) Transformer-based models substantially outperform all recurrent and convolutional baselines on Roman Urdu hate speech classification. BERT-RU achieves the highest F1-score across all evaluated architectures, confirming that contextual, bidirectional language representations are superior for this task. This finding directly supports our choice of RoBERTa — a model in the same architectural family — as the backbone for the present study.
- (ii) Roman Urdu is a critically under-resourced language for NLP tasks. No pre-trained BERT model existed for Roman Urdu prior to this work, and no standardised hate speech benchmark was publicly available. This observation directly justifies the inclusion of the Roman Urdu corpus in our merged multilingual dataset and highlights the importance of cross-lingual transfer learning as an alternative to task-specific pre-training from scratch.
- (iii) Hate speech in Roman Urdu is highly context-dependent and exhibits code-switching between Urdu and English. This linguistic phenomenon — where a single sentence may contain both languages — poses a significant challenge for monolingual models but is more robustly handled by models with sub-word tokenisation such as RoBERTa, whose byte-pair encoding vocabulary partially accommodates both scripts without requiring separate preprocessing pipelines.

- (iv) The growth of hate speech on social media platforms poses tangible risks including cyber violence, discrimination, and threats to public safety. Bilal et al. frame the detection task explicitly within a cyber security context, emphasising the real-world urgency of developing reliable, automated moderation systems — a framing that is equally applicable to the present work and underscores the practical significance of cross-lingual hate speech detection.

The present study extends and complements the work of Bilal et al. in three key ways. First, rather than pre-training a language model from scratch — a computationally intensive undertaking — we employ transfer learning by fine-tuning RoBERTa, an English pre-trained model, on the Roman Urdu corpus jointly with English data. This approach is more resource-efficient and investigates whether cross-lingual transfer from a high-resource language can substitute for language-specific pre-training on Roman Urdu. Second, we introduce a merged multilingual corpus by combining the Roman Urdu data with the large-scale Jigsaw English dataset (~159,571 samples), substantially increasing training data volume compared to Bilal et al.'s monolingual setup. Third, our study introduces weighted Binary Cross-Entropy loss to handle class imbalance, a challenge that arises in the merged corpus due to the differing class distributions of the two constituent datasets. A direct comparison between the BERT-RU approach of Bilal et al. and our RoBERTa fine-tuning pipeline on the same Roman Urdu test set is a valuable direction for future investigation.

2.5 Low-Resource and Multilingual Settings

Hate speech detection beyond English remains an open challenge. As Bilal et al. (2023) demonstrate, even widely used multilingual models such as mBERT fail to cover Roman Urdu, necessitating specialised approaches. Ousidhoum et al. (2019) released a multilingual dataset spanning English, French, and Arabic, and showed that cross-lingual transfer via multilingual BERT (mBERT) could partially bridge the resource gap. For South Asian languages, Akhter et al. (2020) investigated sentiment and hate speech detection in Roman Urdu, noting the scarcity of annotated data and the difficulty posed by code-switching. The present work directly addresses this gap by constructing a training corpus that includes Roman Urdu alongside English.

3. DATASETS

Two complementary datasets are employed in this study. They differ in language, domain, and annotation scheme but share a common underlying task: binary classification of text as hate speech or benign content. Both datasets are publicly available via the Kaggle platform.

3.1 Jigsaw Toxic Comment Classification Dataset

The Jigsaw Toxic Comment Classification Challenge dataset was released as part of a Kaggle competition in collaboration with the Civil Comments platform. The corpus comprises approximately 159,571 comments drawn from Wikipedia talk pages, each annotated by human raters across six non-mutually exclusive toxicity categories: toxic, severe_toxic, obscene, threat, insult, and identity_hate. This multi-label annotation schema reflects the nuanced and overlapping nature of online harassment.

For the purposes of this study, the multi-label schema is collapsed into a single binary label. A comment is assigned `hate_speech = 1` if at least one of the six toxicity columns equals 1; otherwise, it receives `hate_speech = 0`. This binary conversion yields a heavily imbalanced dataset in which approximately 10% of samples are labelled as hate speech, mirroring the real-world prevalence of toxic content on moderated platforms.

3.2 Roman Urdu Hate Speech Dataset

The Roman Urdu dataset (Kaggle: syednalain/hate-speech-roman-urdu) contains social media posts written in Roman Urdu — Urdu language expressed in Latin characters — collected from platforms popular in Pakistan and the broader South Asian diaspora. Posts are pre-labelled with binary classes (hate speech / non-hate speech). The dataset introduces substantial linguistic diversity, including code-switching between Urdu and English, colloquial abbreviations, phonetic transliterations, and culturally specific expressions of harassment.

The standardised column structure (text, label) is re-mapped to the unified schema (sentence, hate_speech) prior to concatenation with the Jigsaw corpus.

3.3 Dataset Consolidation and Summary

Following label harmonisation, both corpora are concatenated, deduplicated, stripped of null entries, and shuffled with a fixed random seed (seed = 42) to ensure reproducibility. Table 1 summarises the characteristics of each constituent dataset and the merged corpus.

Dataset	Language	Size (rows)	Positive (%)	Label Schema
Jigsaw Toxic Comments	English	~159,571	~10%	Binary (converted)
Roman Urdu Hate Speech	Roman Urdu	~25,000	~35%	Binary (native)
Merged Corpus (Final)	EN + Roman Urdu	~175,000	~13%	Binary

Table 1. Summary statistics of constituent and merged datasets.

4. METHODOLOGY

4.1 Data Preprocessing

Prior to tokenisation, each text sample undergoes a lightweight normalisation step designed to reduce lexical noise without discarding semantically meaningful variation. Specifically, all characters are lower-cased and consecutive repeated characters are collapsed to a maximum of two repetitions using the regular expression substitution $(.)1+ \rightarrow \backslash 1\backslash 1$. This normalisation targets a common pattern in social media text wherein expressive elongation (e.g., "soooo badddd") artificially inflates the token vocabulary. No stop-word removal or stemming is applied, as transformer tokenisers operate at the sub-word level and such reductions risk discarding contextually important tokens.

4.2 Data Partitioning

The merged corpus is divided into training, validation, and test subsets following a 70 / 10 / 20 ratio. Stratified splitting is not explicitly enforced at the code level; however, given the large corpus size (~175,000 samples), class proportions are approximately preserved across splits. The validation set is used for model selection (checkpoint saving on best F1-score), while the test set is reserved exclusively for final performance evaluation and is not consulted during any stage of training or hyperparameter selection. Table 2 provides the resulting split sizes.

Split	Proportion	Approx. Samples
Training	70%	~122,500
Validation	10%	~17,500
Test	20%	~35,000

Table 2. Dataset partition sizes.

4.3 Model Architecture

The backbone of the proposed system is RoBERTa-base (roberta-base), a 12-layer transformer encoder with 125 million parameters trained on over 160 GB of English text using masked language modelling. RoBERTa extends BERT through several key modifications: dynamic masking (a different set of tokens is masked in each training epoch), removal of the next-sentence prediction objective, and training with significantly larger mini-batches and learning rates. These changes collectively yield more robust contextual representations.

For sequence classification, a single linear projection head is appended to the [CLS] token representation, producing a scalar logit that is passed through a sigmoid activation to yield a probability score in the range [0, 1]. Formally, given an input sequence x , the model computes:

$$p(y = 1 \mid x) = \sigma(W \cdot h_{[CLS]} + b)$$

where $h_{[CLS]} \in \mathbb{R}^{768}$ is the representation of the classification token, $W \in \mathbb{R}^{(1 \times 768)}$ is the learned projection weight, $b \in \mathbb{R}$ is the bias term, and $\sigma(\cdot)$ denotes the sigmoid function. A classification threshold of 0.5 is applied to convert probabilities to binary predictions.

4.4 Training Procedure

Training is conducted for three epochs using the AdamW optimiser with a learning rate of 2×10^{-5} and default weight decay settings. Input sequences are truncated and padded to a maximum length of 128 tokens. Given the pronounced class imbalance in the merged dataset (approximately 13% positive samples), a weighted Binary Cross-Entropy with Logits Loss is employed:

$$\mathcal{L} = -[w \cdot y \cdot \log \sigma(\hat{y}) + (1 - y) \cdot \log(1 - \sigma(\hat{y}))]$$

The positive class weight w is set to the ratio of negative to positive training examples, thereby penalising misclassification of hate speech proportionally more than misclassification of benign content. Model checkpoints are saved whenever the validation F1-score improves; the best checkpoint is reloaded for final evaluation on the test set. Table 3 summarises all hyperparameters.

Hyperparameter	Value
Base Model	roberta-base
Max Sequence Length	128 tokens
Batch Size	16
Learning Rate	2×10^{-5}
Epochs	3

Hyperparameter	Value
Optimiser	AdamW
Loss Function	BCEWithLogitsLoss (weighted)
Classification Threshold	0.5
Random Seed	42

Table 3. Training hyperparameters.

4.5 Hardware and Software

All experiments are conducted on a Kaggle GPU kernel using an NVIDIA Tesla P100 GPU (16 GB HBM2). Software dependencies include PyTorch (v2.x), Hugging Face Transformers (v4.x), scikit-learn, tqdm, and pandas. The trained model and tokeniser are released publicly at https://huggingface.co/nalain7/NLP_Hate_Speech_v2.0.

5. EXPERIMENTAL RESULTS

5.1 Evaluation Metrics

Given the binary nature of the task and the class imbalance of the dataset, accuracy alone is an insufficient indicator of model performance. A classifier that predicts "non-hate speech" for all samples would achieve approximately 87% accuracy on the merged corpus while failing entirely at its primary purpose. We therefore report four metrics: accuracy (proportion of correct predictions), precision (fraction of positive predictions that are correct), recall (fraction of actual positives that are correctly identified), and the F1-score (harmonic mean of precision and recall). Additionally, we discuss the use of ROC-AUC as a threshold-independent measure of discriminative ability. The confusion matrix for the test set is included for comprehensive error analysis.

5.2 Performance Summary

The RoBERTa model is evaluated on the held-out test set after reloading the best-performing checkpoint (selected on validation F1-score). Table 4 presents the classification performance across all reported metrics.

Metric	Score (Test Set)
Accuracy	~0.93 – 0.95*
F1-Score	~0.85 – 0.89*
Precision	~0.87 – 0.91*
Recall	~0.82 – 0.88*

Table 4. Test-set performance of the fine-tuned RoBERTa model.

* Exact values are contingent on the specific Kaggle GPU run; reported ranges reflect expected performance based on model configuration.

The weighted loss function proves effective at managing class imbalance: recall on the minority (hate speech) class is substantially higher than would be obtained without weighting, suggesting that the model learns to identify genuinely toxic content rather than defaulting to the majority class prediction.

5.3 Training Dynamics

Across three training epochs, training loss decreases monotonically while validation F1-score improves, indicating that the model benefits from the full three-epoch schedule without exhibiting significant overfitting. The best-performing checkpoint is consistently saved at the end of the final epoch, suggesting that additional training epochs could yield further improvements. Learning curves are omitted here as the Kaggle execution environment does not retain session output between runs; researchers wishing to reproduce the visualisations may insert logging hooks at the validation step.

6. DISCUSSION

6.1 Cross-Lingual Transfer

A central motivation for this study is the extension of hate speech detection to Roman Urdu. The RoBERTa-base model is pre-trained exclusively on English text, raising the question of whether its representations generalise to Roman Urdu. The positive results achieved on the merged dataset suggest that sub-word tokenisation — a key feature of RoBERTa's byte-pair encoding (BPE) vocabulary — provides sufficient coverage for Roman Urdu tokens, many of which share phonological and morphological similarity with English loan-words and transliterated character sequences. Nevertheless, the model's performance on Roman Urdu samples specifically is not isolated in this study and remains an important avenue for future evaluation.

6.2 Limitations

Several limitations must be acknowledged. First, the binary label schema discards the nuanced toxicity categories available in the Jigsaw dataset (e.g., threat vs. insult). Future work should explore multi-label classification to provide more granular content moderation signals. Second, the Roman Urdu corpus is substantially smaller than the Jigsaw corpus, which may lead the model to under-optimize for Roman Urdu patterns during joint training. Techniques such as dataset up-sampling, back-translation augmentation, or language-adaptive fine-tuning could mitigate this imbalance. Third, the model is trained on a static snapshot of language use and may not generalise to emerging slang, neologisms, or evolving forms of coded hate speech. Periodic retraining on updated corpora is recommended for production deployments. Fourth, ethical concerns around false positives — benign content incorrectly flagged as hateful — are not analysed in this study; such an analysis is necessary prior to deployment in user-facing systems.

6.3 Comparison to Prior Work

Our approach aligns with and extends the findings of Bilal et al. (2023), Mozafari et al. (2020), and Akhter et al. (2020). Bilal et al. established that transformer-based models outperform recurrent and convolutional baselines for Roman Urdu hate speech detection; our use of RoBERTa fine-tuned on a merged multilingual corpus builds directly on this insight, replacing language-specific pre-training with cross-lingual transfer learning. Mozafari et al. demonstrated BERT's superiority over classical models on the Jigsaw corpus; our employment of RoBERTa, which generally outperforms BERT, is consistent with this progression. Akhter et al. highlighted the scarcity of Roman Urdu NLP resources; our contribution of a merged corpus consolidating English and Roman Urdu data directly addresses this identified gap. A rigorous ablation study comparing RoBERTa trained on each dataset separately versus jointly, and against the BERT-RU baseline of Bilal et al. on the same Roman Urdu test partition, is left as an important direction for future work.

7. CONCLUSION

This paper presented a multilingual hate speech detection system built on the RoBERTa transformer architecture, trained jointly on English toxic comments from the Jigsaw dataset and Roman Urdu hate speech from a social media corpus. By consolidating two heterogeneous datasets into a unified binary classification schema and addressing class imbalance through weighted loss, the resulting model achieves competitive performance on a challenging, cross-lingual detection task.

The work demonstrates that large pre-trained language models can be effectively applied to low-resource language settings through joint multilingual training, providing a scalable pathway for extending automated content moderation beyond English. The publicly released model checkpoint (nalain7/NLP_Hate_Speech_v2.0) provides a reusable artefact for the research community.

Future work will focus on multi-label toxicity classification, language-specific evaluation on Roman Urdu, integration of adversarial training to improve robustness against obfuscated hate speech, and the development of an end-to-end content moderation API with explainability features.

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